

# 3 DOLLARS OF TRASH BAGS AND LEAVES VS A SOAKED BAG: EVERY NUMBER

A wet sleeping bag stops warming you. A bed of dry leaves inside trash bags slows your own body heat enough to reach morning on a cool night. This is a stopgap, not a sleeping bag with a rating.

## WHAT IT IS

Two trash bags stuffed with dry leaves lie on a loose foot of leaves. A third bag goes around your legs, and another foot of leaves goes over you on a stick frame. Your body makes the heat. The leaves just slow how fast it escapes by trapping still air.

On a cool night that bed leaks about 21 watts on paper, against the 70 to 80 watts your body makes while you sleep. The margin looks wide on paper and gets thin in real weather. Nobody has tested this exact bed.

## WHY THE SOAKED BAG FAILS

|            |                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------|
| above 40F  | Hypothermia does not need freezing. Rain, sweat, or a soaked bag can start it at temperatures above 40 degrees Fahrenheit. |
| 70 to 80 W | Heat your body makes while you sleep. A bag makes none of its own, it only holds still air to slow the escape.             |
| about 23x  | Water conducts heat about 23 times better than still air. Once water fills the bag, your heat has a fast way out.          |

Wet down collapses and loses loft. A wrung out synthetic bag keeps some warmth, but a fully soaked bag is still a bad bet for the night.

## WHY LEAVES WORK: THE MATH

|            |                                                                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 to 16 mm | German researchers shredded autumn leaves into pieces this size and poured them loose.                                                                                   |
| about 60%  | That loose pile carried about 60 percent more heat than still air at room temperature and middling humidity. Same range as cellulose and wood fiber insulation.          |
| doubled    | Your leaves are not shredded and you lie on them, so the video doubles that number. It is a guess, not a measurement, and it is the planning number for everything here. |

Worked on a cool night near 37 degrees Fahrenheit, with skin near 91 degrees, a gap of 30 degrees Celsius.

|               |                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| about 8 W/m2  | Heat a foot of leaves on top passes through each square meter.                                                       |
| about 12 W/m2 | Heat the squashed base passes if it settles to 8 inches. More than half the loss goes down, so the base comes first. |
| about 2 m2    | Your skin area, roughly half facing up and half facing down.                                                         |
| about 21 W    | Whole bed loss on paper, against the 70 to 80 you make.                                                              |
| about 62 W    | If the real loss is three times the paper number. That is barely under what you make, so the margin is thin.         |

Thickness is the lever: double it and the heat flow drops by half. If you run short of leaves, run short on top, never underneath.

FIRST FIFTEEN MINUTES

- 1 If you have a signal, call for help and say exactly where you are.
- 2 Get out of the wind. A rock face, a log, or the downwind side of a hill is enough for now.
- 3 If you are alert and have food, eat something. Your body burns fuel to make heat.
- 4 Decide. A building or a car with working heat within easy reach beats the bed. Go there now, while your hands still work.
- 5 If not, start gathering while you are still warm. The work goes better warm.

GATHERING THE LEAVES

The bed wants a foot of loose leaves under you, two bags stuffed full, and a foot over you. That is roughly 260 gallons, more than six bags worth. Start before dark.

- ☐ Use the bags as haulers: stuff, drag, empty, repeat. Check each for holes before the final fill.
- ☐ Gather the driest leaves you can find, in the lee of a wall or log and under thick evergreen boughs.
- ☐ Squeeze a handful. Dry leaves crackle and spring back, damp ones clump. Skip anything flat and matted on wet ground.
- ☐ Dry pine needles and dry grass work too. After rain, take what you find.
- ☐ Keep leaves loose and fluffy. Packing squeezes out the air that insulates.
- ☐ Skip any leaf or vine you cannot identify, dead ones too. Poison ivy oil sticks to surfaces for years.
- ☐ Gather more than you think you need. Even dry leaves pull water from humid night air.

BUILDING THE BED

- 1 Pick a spot out of the wind, a log or rock face on the windward side. Look up for dead branches, stay out of low spots where water and cold air pool.
- 2 Clear sharp sticks and stones, but leave the forest floor, it already insulates.
- 3 Check each bag for holes. Lay a base: a loose foot of dry leaves the length of your body, a little wider than your shoulders. A sleeping pad goes on top of the base if you carry one.
- 4 Stuff two bags into fat cushions. Fold each opening over twice and tuck it under, openings on the dry side.
- 5 Lay them end to end, one under shoulders and hips, one under legs. Lie on it. If you feel the ground, add leaves until you cannot.
- 6 Lean long sticks from the windward log over the bed about a foot apart. Weave small sticks across. Push down to test the ribs before you trust them.
- 7 Pile leaves over the frame at least a foot thick, more if you have time. Lay light branches on top so wind cannot strip it. Leave the head end open.
- 8 Put on every dry layer, loose enough not to sweat. Pull a hat down, wrap your neck, keep it off your mouth and throat.
- 9 Sit at the open end, legs inside. Feet into the third bag, pull it up as far as it reaches. Fold the top edge well below your chin, arms and shoulders outside the plastic.
- 10 Lie back. Pull leaves around shoulders and neck to close the gap. Keep open space in front of your face and make sure you can push out through the pile.

WHAT THE PLASTIC IS FOR

Plastic adds almost no insulation. The leaves insulate. The plastic blocks wind, keeps ground damp out of the bottom leaves, and keeps wet clothing off dry leaves.

|              |                                                                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| nearly 10C   | A lab dummy filled with warm water cooled nearly 10 degrees Celsius in a bit over two hours inside a plain plastic survival bag.                         |
| just over 3C | The best insulated system in that test lost just over 3 degrees. Insulation, not plastic, does the work.                                                 |
| 19 to 42%    | A Swedish test on a dummy in wet clothes: a plastic vapor barrier cut total heat loss by 19 to 42 percent, about the same as taking the wet clothes off. |

WHAT IT COSTS

about 30 dollars

A box of 50 contractor bags in the 42 gallon size at a big hardware store. That is about 60 cents a bag.

under 2 dollars

The three bags this bed actually uses. The catch is you buy the whole box, so the real price is about 30 dollars.

47 spares

The rest of the box. Keep three rolled up in your pack or car, where they take almost no room.

33 by 45 in

The third bag laid flat. On most adults it reaches the waist or the ribs.

That price does not cover a roof. If you carry a tarp or poncho, rain is handled. If not, add one to your pack list. A roof decides whether this bed survives a wet night.

THE SOAKED BAG AND WET CLOTHES

- ☐ Wring the sleeping bag out as hard as you can. Drape it over your legs, outside the plastic. Wet things go outside the plastic, never against your clothes.
- ☐ Once in the bed and out of the wind, swap wet layers for dry ones, one piece at a time so you are never bare.
- ☐ That swap is worth the cold minute it costs. Wet clothes pull heat all night, dry ones do not.
- ☐ Nothing dry? Wring out what you can and keep it on.

THROUGH THE NIGHT

- ☐ If you start to sweat, loosen a layer or open the top of the bag until you cool, then close it. Go by how you feel, not a timer. Sweat inside plastic soaks your clothes.
- ☐ A little shivering when you first settle is normal. It is your body making extra heat.
- ☐ If it turns hard, add leaves where you feel drafts and check the ground under you.

DANGER SIGNS: THIS BED IS NOT ENOUGH

- ☐ Hard shivering that will not settle, or any confusion, slurred speech, or clumsiness.
- ☐ Shivering that stops while you feel worse is a danger sign, not a good one.
- ☐ Call emergency services if you have a signal, and follow their instructions on whether to move or stay put.

NO SIGNAL

Make noise and light anyway. Three of anything is the distress signal: blow three blasts on a whistle or flash a light three times, then wait and repeat.

TEST IT BEFORE YOU NEED IT

Build this once in daylight in your backyard. It is a fit test, not a cold test. Learn how many bags of leaves it takes, how long gathering runs, and whether you feel the ground. Lie in it for an hour and see if gaps open at your neck and feet.

## WHERE THIS STOPS WORKING

Everything here assumes a cool night near 37 degrees Fahrenheit, still air, and dry leaves. A sleeping bag has a temperature rating. This bed has none.

### about 29 W

Paper loss at 15 degrees Fahrenheit. Three times that comes to 88 watts, more than you make.

### near freezing

The tripled number already sits right at what you make. Below freezing, this is a stopgap to reach morning, not a plan.

**Rain hurts it badly.** Steady rain soaks a leaf pile from above and wet leaves insulate worse. In rain, put a roof up first: a tarp, tent fly, or poncho over the frame, sloped so water runs off, open end away from the wind. No tarp in steady rain means this bed will not hold, and any real shelter beats it.

## FACE AND FLAME

- ☐ **Keep your face and nose clear all night.** Never pull the bag or the leaves over your face.
- ☐ **Keep any flame far from the bed.** No candle, stove, or lantern. Dry leaves burn fast and plastic melts.
- ☐ Nothing here treats serious hypothermia.

## WHO SHOULD SKIP IT

- ☐ **Infants and small children do not sleep in this build.** Plastic bags are a suffocation hazard for infants. For a child, the plan is warm shelter and a call for help.
- ☐ Anyone who cannot get out of the bag alone, or is too weak or confused to sit up, needs rescue, not a leaf bed. While help comes, keep them lying flat, handle them gently, and give no food or drink unless they are fully alert.
- ☐ This bed fits one adult. If two people are stuck, build two beds side by side.

## IN THE MORNING

- Do not rush out at first light. Eat, drink, and get warm in the bed first.
- Wring out the sleeping bag again and spread it in any sun or wind.
- If you called for help, stay where you told them. If not, move toward shelter in daylight, slow enough that you do not sweat.
- Keep the trash bags. They still work as a wind and rain shell.

Same body, same 70 to 80 watts, different problem next: a cold bedroom in a blackout, and what a 26 dollar foam pad under you does. Watch "When the Power Goes Out, Your Bedroom Is Too Big" on the channel.

**Watch the full video:** [youtube.com/@SketchySurvival101](https://youtube.com/@SketchySurvival101)

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Subscribe so the next soaked bag does not catch you flat footed.

Sketchy Survival 101. Physics first survival. Every number on this sheet comes straight from the video, with its hedges kept: doubled planning guess, paper versus real loss, dated and foreign lab estimates, and typical values that are not safety limits. This is a stopgap for a cool night, not a rated sleeping bag. When in doubt, call for help and get to real shelter.